

Umut Kocasari

PhD Researcher in 3D Computer Vision

Email | LinkedIn | GitHub | Google Scholar | Website

Research Profile

PhD researcher in 3D computer vision at the Technical University of Munich, working on 3D/4D reconstruction and geometric deep learning. First author of an ECCV 2026 Oral (Spotlight) paper on 4D face reconstruction. Research experience spans geometry reconstruction, neural fields, 3D style transfer, and latent-space manipulation.

Education

Technical University of Munich (TUM)

Ph.D. in Informatics, Visual Computing Group

Oct 2025 – Present

Munich, Germany

- Supervised by Prof. Matthias Nießner.
- Research interests: 3D reconstruction and geometric deep learning.
- Expected completion: Oct 2029.

Technical University of Munich (TUM)

M.Sc. in Informatics

Oct 2022 – Jul 2025

Munich, Germany

- Grade: **1.3** (German grading scale; 1.0 = best).
- Research in neural fields, 3D reconstruction, and geometric deep learning.

Boğaziçi University

B.Sc. in Computer Engineering

Sep 2016 – Feb 2022

Istanbul, Turkey

- GPA: **3.78 / 4.00**.
- Research in computer vision and latent-space manipulation in GANs.

Publications

Umut Kocasari, Simon Giebenhain, Richard Shaw, and Matthias Nießner. “*Face Anything: 4D Face Reconstruction from Any Image Sequence.*” European Conference on Computer Vision (ECCV), **Oral (Spotlight)**, 2026.

Adil Meric, **Umut Kocasari**, Matthias Nießner, and Barbara Roessle. “*G3DST: Generalizing 3D Style Transfer with Neural Radiance Fields across Scenes and Styles.*” German Conference on Pattern Recognition (GCPR), 2024.

Enis Simsar, **Umut Kocasari**, Ezgi Gulperi Er, and Pinar Yanardag. “*Fantastic Style Channels and Where to Find Them: A Submodular Framework for Discovering Diverse Directions in GANs.*” Winter Conference on Applications of Computer Vision (WACV), 2023.

Berkay Doner*, Elif Sema Balcioglu*, Merve Rabia Barin*, **Umut Kocasari**, Mert Tiftikci, and Pinar Yanardag. “*PaintInStyle: One-Shot Discovery of Interpretable Directions by Painting.*” Computer Vision for Fashion, Art, and Design (CVPR Workshop), 2022.

Umut Kocasari, Kerem Zaman, Mert Tiftikci, Enis Simsar, and Pinar Yanardag. “*Rank in Style: A Ranking-based Approach to Find Interpretable Directions.*” Computer Vision for Fashion, Art, and Design (CVPR Workshop), 2022.

Umut Kocasari, Alara Dirik, Mert Tiftikci, and Pinar Yanardag. “*StyleMC: Multi-Channel Based Fast Text-Guided Image Generation and Manipulation.*” Winter Conference on Applications of Computer Vision (WACV), 2022.

Umut Kocasari*, Alperen Bag*, Efehan Atici, and Pinar Yanardag. “*Exploring Latent Dimensions of Crowd-sourced Creativity.*” Machine Learning for Creativity and Design (NeurIPS Workshop), 2021.

Professional Experience

Siemens

Deep Learning Working Student

Aug 2024 – Aug 2025

Munich, Germany

- Developed and improved deep-learning models for real-time emotion and gesture recognition from webcam video.
- Improved the performance and accuracy of existing detection models.
- Implemented an icon-search system based on semantic similarity.
- Built an interactive tool for icon creation and editing.

Hohenstein

Machine Learning Working Student

Feb 2024 – Aug 2024

Munich, Germany

- Developed a machine-learning system to predict appropriate test parameters for different materials.

- Built a GUI for visualizing model predictions and improving model explainability.
- Created a decision-support system that reduced the workload of product specialists by **20%**.

Armut

Jul 2021 – Feb 2024

Data Scientist

Istanbul, Turkey

- Developed a job-fulfillment prediction model that increased fulfillment rate by **5%**.
- Implemented a contrastive-learning-based matching algorithm that improved quote rate by **30%** over previous methods.
- Built a customer-satisfaction prediction model achieving **77% accuracy**.
- Developed an NPS simulation model with a mean absolute error of **5 points**.
- Improved an identity-card validation pipeline using deep-learning methods, achieving **80% validation accuracy**.

Eczacıbaşı Bilişim

Nov 2020 – Jun 2021

Data Analytics Intern

Istanbul, Turkey

- Developed a demand-forecasting system that reduced MAPE by **15%** compared with previous methods.
- Worked on sales-force optimization using field data.

Boğaziçi University

Oct 2020 – Feb 2021

Teaching Assistant, Department of Computer Engineering

Istanbul, Turkey

- Teaching assistant for *CMPE 250: Data Structures and Algorithms*.
- Led tutorials, contributed to course projects, and graded student work.

Allianz

Jul 2020 – Sep 2020

Information Technology Intern

Istanbul, Turkey

- Implemented OAuth 2.0 authentication in collaboration with the software team.
- Investigated blockchain technology and presented findings to senior management.

Artiwise

Jul 2019 – Sep 2019

Data Science Intern

Istanbul, Turkey

- Developed a neural model for spelling correction and text normalization.
- Evaluated the system on a sentiment-analysis pipeline, improving downstream accuracy by **1%**.

Technical Skills

Research Areas:	3D/4D Reconstruction, 3D/4D Face Avatars, Geometric Deep Learning, Computer Vision, Neural Fields, 3D Style Transfer, Latent-Space Manipulation
Programming & Frameworks:	Python, C++, PyTorch
Additional ML:	Deep Learning, Machine Learning, Predictive Modeling, Natural Language Processing

Selected Awards & Honors

- **1st Place**, G-Research Competition — 12 teams, 2023.
- **1st Place**, Datathon 2022 — 165 teams, 2022.
- **1st Place**, Kariyer.net Datathon — 26 participants, 2021.
- **1st Place**, Techcareer Datathon — 52 participants, 2021.
- **1st Place**, Trendyol Datathon — 8 teams, 2020.
- **3rd Place**, Insider AI Race — 60 participants, 2020.
- Ranked **152nd among 2.2 million students** in the 2016 Turkish National University Entrance Examination.

Languages & Interests

Languages: Turkish (native), English (C1 Advanced)

Interests: Chess, volleyball, reading